

“I knew Open Source hardware was a necessity to redesign into newer hardware”



■ MARCUS ERLANDSSON

Marcus Erlandsson has been in the semiconductor business since 1998. He started as a field-programmable gate array (FPGA) and application-specific integrated circuit (ASIC) design consultant.

In 2000, he moved into the ASIC development business, and one of the key focus areas of the company he worked for was FPGA-to-ASIC and ASIC-to-ASIC conversion. It was during this period that he identified and began to participate deeply in OpenCores—a Web community which hosts a repository of free, Open Source intellectual property (IP) cores including chip designs, system-on-a-chip and supplemental boards. Erlandsson's involvement in OpenCores became so deep that it eventually became a part of ORSoC (OpenRISC System-on-Chip)—an FPGA/ASIC design house that he founded with two other colleagues in 2004.

We caught up with Erlandsson on the significance of Open hardware, the OpenCores journey, the business case behind it, how it is benefiting engineers worldwide, and more business and technical doubts that commonly crop up...

What caused the birth of OpenCores.org?

OpenCores was started by a Slovenian engineer in 1999. The goal was to design the world's first Open Source, 32-bit, reduced instruction set computer (RISC) processor licenced under the Lesser General Public Licence (LGPL).

There were just five to ten engineers initially. Today, OpenCores has around 850 maintainers taking care of around 770 projects, and the number is growing rapidly.

Do you notice any significant change in the outlook towards Open Source hardware in recent times? Correspondingly, has the growth of OpenCores.org also accelerated in recent years?

Yes, I have been promoting OpenCores and Open Source hardware since 2001 and the acceptance level has really changed since 2006-07. Open Source hardware has got a lot of help from Linux. Linux introduced Open Source into commercial companies and this has helped OpenCores to get acceptance. Also, the fact that ORSoC (a professional design

company) is standing behind OpenCores has helped to raise the status of Open Source hardware, as there is the assurance that there is a company offering professional support related to this technology.

Is OpenCores.org for-profit or not-for-profit?

I would say a mix of the two. We offer banner advertising on the site, and it has become quite popular amongst global companies within the electronics business. OpenCores also offers a service to help companies find suitable resources, for employment or as consultants. We try to cover the existing costs of running and maintaining the OpenCores servers.

Do you charge for the projects on OpenCores.org?

We do not charge anything from our users and there are no costs involved in starting a project at OpenCores. It is also totally free to download and use the different cores available.